



WORLDS OF **SCIENCE FICTION**

Returned from the stars . . .

MAN ALONE

By DON BERRY

OCTOBER
35 CENTS



IND



WORLDS OF SCIENCE FICTION

Returned from the stars . . .

MAN ALONE

By DON BERRY

OCTOBER
35 CENTS



IND

The Project Gutenberg eBook of A Toothache on Zenob

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A Toothache on Zenob

Author: Boyd Ellanby

Illustrator: Ed Emshwiller

Release date: October 23, 2019 [eBook #59665]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59665

Credits: Produced by Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK A TOOTHACHE
ON ZENOB ***

A TOOTHACHE ON ZENOB

BY BOYD ELLANBEE

*Strange to think that from twenty-odd
light-years away, other eyes see our own Sun
blazing in the middle of a familiar constellation....*

**[Transcriber's Note: This etext was produced from
Worlds of If Science Fiction, October 1958.
Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

Pehn Karn sat in the signal dome, idly waiting while his friend, adjusted the dials of the receiver. The recording tape spilled over the table in loops of aluminum.

"Doesn't this job get dull?" he inquired.

Nautunal turned the fifth dial a few degrees, and glanced up. "Depends on your interest. It's true this is just routine space-sweep, but noise from space is amazingly variegated. Just one more sector to scan tonight, and I'll be through. What's matter with you, Pehn? Your face looks little lopsided."

Pehn tried to grin, and fingered the slight swelling on his cheek. "My face will never be my fortune, I guess. I don't know what's matter. Just ache."

Even at his best, as Pehn's family loved to remind him, he was an ugly young man; he had none of the rounded placidity of feature which was the ideal of his race. His olive skin stretched too tightly over his cheekbones, and his black eyes peered too intensely from their deep sockets. It helped very little that he happened to be extremely intelligent.

Pehn covered the aching place with his hand, and tried to concentrate on the emerging spills of tape. As a matter of fact, he had been having severe periodic toothaches for six months now, but had never spoken of it.

Suddenly he bent forward. "Hold it! Just minute. Let me see that."

Nautunal raised one eyebrow. "Don't let it get you, lad. Listening to space is apt to make you jumpy. Your friends over at Atomics wouldn't like that. More than once I've thought I was finding some sort of sense in all this chatter, but it never pans out. It's just noise. There may be other inhabited planets besides Zenob, just as Bidagha claims, but if so, they aren't talking."

"Stop, watch tape," said Pehn. Nautunal shrugged his shoulders, but he picked up the tape and watched as it trickled through his fingers.

The machine was recording short bursts of energy, separated by distinct pauses: ".. .. ."

"Two and two," remarked Pehn, "are four. Three and three are six. Four—"

"I know. I've been to kindergarten too. Four and four are eight. Has your aching face affected your mind? You ought to submit yourself for treatment."

He reached to shift the scanner, but Pehn grabbed his hand.

"Can't you see? Somebody is trying to show us they know how to add. Someone out in space. Keep watching. I wonder if they use duodecimal system, or what? Where is it coming from?"

Nautunal dropped his skeptical pose, and watched the emerging tape in silence. The growing table of symbols built all the simple additions up to 10 plus 10, by the laborious accumulation of dots. Then it began again, systematically, "One and one are two, two and two—"

Pehn turned his wondering eyes on his friend. "Is this trick? Joke you rigged up for my benefit?"

A. D. 2120:



"Is this a trick?"

Nautunal shook his head, and his voice was hardly a whisper. "No. That stuff is really coming through space—through phase space."

Twice more, the table of additions appeared. Then, after a brief pause, came simple multiplications. Hour after hour the signals continued, endlessly repeated, and shortly after midnight the two could recognize the periodic table of chemical elements, with atomic weights and numbers of isotopes.

"If those numbers which follow atomic weights are abundances," said Pehn, "composition of their planet is not quite same as ours. Look how rare 235 is. Where do you suppose this is coming from?"

"You guess," said Nautunal. He waved his arm towards the transparent dome through which the stars shone, and grinned. "From little data I have so far, signals might be coming from somewhere in Weaver, perhaps from neighborhood of Topaz, but it's too early to be sure of anything."

Suddenly a marked change occurred in the pattern of the signals. The clear symmetry of mathematics ended and was replaced by a formless jumble, but a jumble whose repetition suggested that it, too, contained a pattern if it could once be glimpsed. Meters and meters of tape piled up, and the young men stared at it in frustration.

Nautunal stood up in sudden decision. "Bidagha is right. There must be intelligent life in another part of galaxy. We need help, Pehn. We can't decipher this stuff, and yet it may be key to basic vocabulary. We need mathematicians, linguists, semanticists. I'll put out call to director." He lifted a finger to activate the visiphone in his wrist band, but before touching it he glanced at his friend in some concern.

"But you don't look well. Perhaps you ought to go home and get some sleep?"

Pehn shook his head. "No, pain will probably be gone by morning, and at time like this sleep would be only gift from Evil Ones. I'm going to get Bidagha. He'll be more use to us than dozen semanticists."

"Call him on visiphone."

"You know he can't wear one. He's at Cave tonight, holding Ceremony. I'll go after him."

"All right, Pehn. But remember, government will probably disapprove of this business. Whatever you do—don't tell your father!"

Pehn grounded his copter at the outskirts of the city, then turned his back to the glowing lights and walked north across the darkened fields towards Cave. The early morning sky blazed with stars, and ahead of him, low on the northern horizon, gleamed the sprawling constellation of Weaver. He had never been able to force his imagination to see many of the constellations in their completeness, and in the patterns of stars which his pastoral ancestors had conceived as Weaver, Sower, Horned Toth, he could see only random clusters of suns. He watched it now, as he walked over the rutted earth, and suddenly the pattern took shape, so that he could discern

the old lady's Shuttle, and at its tip that brilliant yellow star, Topaz, which might that very moment be sending its signals through the galaxy. How many planets revolved around Topaz?

He stopped, for the field ended in a sharp bluff which descended to a narrow valley. Across the valley's floor was the entrance to Cave. He could see the bobbing lights of candles, down there, and hear a muffled chant of many voices. He hoped Ceremony would end, soon, so that he could consult his friend. Once again he felt impatient that Bidagha should have to be met in person just because, as a Healer, he could not wear a visiphone into Cave. Bidagha was really more progressive than many scientists. But the culture of Zenob still had a strongly anti-materialistic, one might almost say anti-scientific tinge, and no machine of any kind could ever be brought into any of Sacred Places. Cave had been the chief place for the ceremonies of those living in Lahzen area for so many thousands of years, that even the historians did not know of a time when it was not in use. It was so old, some heretics said, that it had outlived its usefulness, and was not even a safe place to be in.

The stars were fading, and the northern sky was paling when the chanting stopped in Cave below. People filed out silently, extinguishing their candles as they reached the opening, and last of all came the Healer.

"Bidagha!" Pehn called softly. "Up here!"

The tall figure paused, then ran lightly up the steps out in the sloping hill.

"Has something happened?"

"Hurry. My copter is waiting back there, and I'll take you to signal dome. Then I'll have to go home. If I'm not there for breakfast, my father will begin another lecture on depravity of youth."

Bidagha's eyes twinkled. "Premier Karn is pretty conscious of his responsibilities to nation, Pehn, but perhaps eighteen years ought to be more respectful of fifty. I am nearly latter myself, you know. But what has happened?"

Pehn raised his arm and pointed towards Weaver. "We're getting signals. We think maybe they come from some planet of Topaz."

Bidagha clasped his strong hands on his breast. His black hair, curling over a high, olive forehead, was held in place by the narrow green band of his calling. Under his little mustache his mouth was firm and serene, and his gray eyes were exalted as he stared at the fading yellow star.

"At last!" he said.

The bluish sun of Zenob had risen by the time Pehn got home. He sat through the family breakfast with his parents and his sister, thankful that his night's absence had apparently not been noticed.

Pehn's father, old Premier Karn, represented, the old man liked to think, the ideal of Zenobian maturity. The placidity of his mind was reflected in the soft roundness of his unlined face, and his full lips curved at the corners in perpetual contentment. Like most of his countrymen, he had never felt any conflict between his own impulses and the customs of his society, and never in his life had he needed the ministrations of a Healer.

In his usual benevolent mood this morning, Premier Karn entertained his family with his meditations on his favorite theme, the glories of Zenob's history and the perfection of her knowledge. They listened to his remarks in patient silence. As he sipped at his last cup of fragrant *akhlai*, he tried to make a kindly remark to his son.

"And what progress is your institute making towards practical atomic power, Pehn? Foundations of this work were laid down more than two hundred years ago, and government would be glad to have demonstration at any time."

"We're little uncertain, sir, when that will be. We want to be sure, before our first trial, that we have checked against even vaguest possibility of starting widespread chain reaction."

The Premier frowned, set down his cup, and touched his napkin to his lips. "But how could this occur?"

"It is not at all likely. But if, for example, as has been suggested, crust of our planet should contain large quantities of some heavy element with

properties we don't know about, something related to 235 or 238 for instance, and easily fissionable, our very first trial might prove disastrous."

"Nonsense, nonsense!" said the Premier. "Complete chemistry of our planet was worked out and tabulated more than three hundred years ago. There were great chemists in those days, and since then no further research has ever been necessary. There could not possibly be any elements which we don't know about. It is not seemly for you young men to be questioning work of great geniuses of past."

Pehn's sister, Soma, had been silent, as befitted a woman. Now she said, "Father, I have been told that long time ago, Sarkar Talat, after life spent in philosophical research, gave Warning to government—"

"Trouble with you, my dear," and the Premier patted his pretty daughter's arm, "is that you don't realize women usually lack spiritual insight necessary to interpret veiled words of Ancients. Of course I admit our Healers can foresee future, but they don't always describe it in unambiguous language. Actually, Sarkar issued *two* Warnings, but they obviously did not mean what they seem to at first glance. First was, that we are in danger because we only think we have mastered all basic knowledge. Second was, that there are undoubtedly other worlds in universe, and one of them will one day affect destiny of Zenob. Taken at face value, these are obviously both false. As to first, no new knowledge has been added to our sciences for generations, in spite of fact that Ainta ages ago showed us how to use faculty of precognition. As to second, it is clearly foolish to think Sarkar meant other worlds in physical sense. He must have meant spiritual worlds." He turned again to Pehn, who had risen from the table and was waiting politely.

"When can we have that demonstration? Within month?"

"I'll speak to director today, sir." The throbbing in his cheek was becoming evident again, and he touched his face gently.

"What's matter, Pehn?" said his mother.

"Nothing. My face hurts, little."

"Ignore it!" his father ordered. "I won't have any maladjustment in Lord Karn's family!" He picked up the gold-headed cane which was his badge of office, and strode out of the room.

Pehn managed to spend a second night with the sleepless group of experts at the signal dome, but fatigue, and the growing pain in his cheek sent him home again just before dawn. Softly he ran up the ramp to the second floor and into his bathroom, to the medicine closet.

In Pehn's family, a transient illness was an embarrassment, a persistent illness a disgrace. It had always been his mother's pride, and his father's boast, that in the Karn household the contents of the medicine shelves were never needed, and that the doors of the cupboard remained closed from one year to the next.

It was with a sense of guilt, then, that Pehn pressed a spot on the green-tiled wall to slide back the cupboard doors, and picked up an ivory box, from which he took a bolus of pain-killing plant extract.

He swallowed the huge pill, then took another. A double dose, this time, for he knew the pain would never yield to anything less. He stood shivering for a few moments, waiting for the drug to take effect. He looked up, and realized that his sister was standing at the open door, watching him sympathetically.

"Pehn," she said, "you're ill. Won't you talk it over with me?"

"I would talk it over with one of Evil Ones if I thought it would help this pain. It grows worse and worse."

"Have you told Father yet? He could arrange for Healer—"

"No!" he shouted. "I don't want him to know. He'd only begin lecture on his shame, his gray hairs, how all pain comes from unruly mind, why don't I put myself in tune with group—old familiar story. But I know this is something different."

"But Pehn! You know yourself, surely, that since you've got into Atomics Institute, you have changed. You aren't perfectly adjusted, any more. You worry about things."

He touched his swollen face and smiled at her placatingly.

"Don't worry, Soma, or you'll get lines in your forehead, and Father could never bear shame of having *two* maladjusted children in family. I'm stronger than you think. Last year when I passed Fire Test, I stood flames longer than any of boys in my class, longer even than Nautunal. But this is

different. Trouble with Zenob is that we don't have any biology or any real medicine."

"Pehn! How can you say such things? Your best friend is Healer."

"You know I haven't anything against Healers. But only ailments they can cure are those that originate in mind, and they can't really do anything for purely physical ailments. Bidagha has admitted as much to me. He wants to change all that—he thinks time is ripe."

"But Pehn, our Healers can foresee future. You know that Ainta Penab proved that, five centuries ago."

"Ainta was worst disaster that ever befell Zenob," shouted Pehn. He put out his hand to restrain his sister, who was attempting to struggle to her feet in horror.

"Oh, Soma, I guess I'm just heretic. But listen to me. Sure Ainta Penab proved conclusively that such phenomena as telepathy and precognition are real. And what was result? Members of Cult used it as argument and launched campaign to stop scientific research completely. They nearly succeeded, too. Medical research has never been resumed on any large scale, and chemistry and physics only in last century."

Soma sighed. "You shouldn't get all excited, Pehn. Come down into living room, and stretch out on couch. I'll rub your forehead. You may be ugly old atavism but you're only brother I have, and I want to keep you."

In the big living room, Soma drew the curtains from the lucite walls that looked over the white city. The early sunlight came in, warm and faintly blue, soothing. She put foam pillows under his aching head, and drew up a footstool beside him. She stroked his forehead and he was beginning to doze, lightly, when a muted chime roused him.

He activated his wrist dial, to find Nautunal's grave face looking at him. Pehn sat up.

"What have you found?"

"Something. Can you come here, now?"

Pehn struggled to his feet. "Just give me minute to clear my head."

"Don't go, Pehn!" cried Soma. "You're not well! It's nearly breakfast time, too. Don't go. What will I tell father?"

But the door had closed, and from the window Soma watched Pehn's copter rise above the rooftops and glide out of sight.

In the signal dome Pehn found the specialists still at work, pale and tense from lack of sleep. Some still conferred over pages of mathematical equations, some watched the tape which monotonously continued to record the symbols.

Bidagha strode forward to meet him.

"We've finally found key to vocabulary, Pehn, and others are at work now on main body of message. Signals are undoubtedly coming from one of planets of sun we call Topaz. It has system of nine planets, and waves are being sent from third. All my life I've believed that there were other planet systems like ours, and other intelligences in galaxy, but now that one of them is signalling us, I can hardly make myself believe it!"

From the top gallery, high in the dome, Nautunal ran down the spiral ramp.

"We're getting it now, Bidagha! Planet of Topaz, is signalling this entire sector of galaxy, hoping to find planets with intelligent life on them. They wish to communicate, to exchange information, and they offer to visit any planet which would welcome them."

"Then they can travel through space?"

"Yes. They say they are only twenty-seven light-years away."

Bidagha's shoulders drooped in disappointment. "But if these signals have been twenty-seven years on way, it would take us another twenty-seven years to answer them. I shall never live to see their visit!"

"But you don't understand, Bidagha. I forgot this isn't your field. These waves are coming through phase space, and they go much faster than speed of light. They reach us almost instantaneously."

Pehn began to laugh. He felt light-headed, and for some reason amused. "I suppose you all remember my father's Jarlu lecture, in which he demonstrated mathematically that life or intelligence on any planet except Zenob was impossible. He pointed out that unique properties of carbon compounds of which we are composed could never be duplicated on another planet. And how remarkable circumstance of our having Ice I, only form of solid water that can float, had allowed life to evolve on Zenob, alone. Poor father! This will be hard on him. Of course, you are not going to send answer?"

Bidagha had seemed to be in meditation, but now he spoke in a commanding tone. "Of course we must reply!"

Nautunal gasped, and stepped away. "But we haven't right! That would be heresy!"

"I claim right. Centuries ago, Healers dreamed of this day, and as Healer I dare to claim right. We will reply to these signals and tell people of Topaz planet that intelligent men do exist on our own world."

Nautunal's eyes had become dreamy and speculative. "I am not certain that we *could* reply, even if we dared," he said. He looked uncertainly at the other microwave technicians, to see fear looking from their faces, too.

At last one spoke. "I think we could change over to transmission through phase space in about half hour—if we had orders." He hastily turned away, afraid of his own thoughts.

"Responsibility is mine," said Bidagha. "I speak with authority of my calling. We will send same set of mathematical tables we have received, and then periodic table of elements as they exist here."

Pehn felt confused, battered with warring emotions, and too tired to think or speak. But Nautunal moved with abrupt decision.

"If you order it, Bidagha, we will try." He turned to his technicians. "We'll start work immediately."

Nearly an hour had passed before the wave modulator was reported ready. In the highest level of the dome, they watched nervously as Nautunal turned up the power and worried the dials.

"All set?" he said.

There was no reply. In a dead silence he touched the button, and started an impulse driving towards the star Topaz.

The door behind them opened with a crash and Premier Karn strode in, his face contorted with an anger he had never shown before in his life.

"Despicable traitors!" he shouted. "Turn off that instrument!"

White-faced, one of the technicians obeyed, and the power indicators dropped.

"Can it be true, what has come to my ears—what I see now with my own eyes—that you would dare to reply to message that comes from foreign planet?"

Bidagha's commanding figure grew even taller.

"Premier, I speak now, not as Bidagha man, but as Bidagha Healer, and I must give you my Warning: Zenob cannot escape contacts with other worlds!

"In my opinion, Zenob has reached fateful turning point in its history. We must face fact that our knowledge of physical science is not adequate. Our fossil fuels are nearly gone, and we must have atomic power. But frankly, our physicists don't know enough to design safe atomic reactors. And at rate science progresses on Zenob, they won't know enough for centuries. All of this could be remedied by exchange of information with this other world."

As he faced Bidagha, the Premier trembled with rage, and his usually placid face was contorted. "Bidagha, you could be unfrocked for that! Zenob would be better off if Healers would confine their opinions and activities to healing, and leave politics to those whose business it is. Government will not base its decisions upon visions of dreamers who have dwelt too long in Cave."

Pehn closed his eyes as a wave of agony broke over him. The voices receded; dimly he was aware that he was falling. Pain and shouting together faded away into darkness.

On Earth, at the listening post on Long Island, Joe Weber sat before his recorder, intent on the noise from space.

The chief anthropologist studied the star map on the wall as he asked his question. "Have you got anything at all, Joe?"

The technician shook his head. "Nothing but noise. I've only been beaming them for four days, and our schedule calls for a week on each sector. I'll keep on as planned, but I'm positive that the star systems in the Lyra sector are not inhabited in any sense we would recognize, or, if there is life there, it hasn't developed enough of a science for them to know they're being signalled."

The anthropologist sighed. "It may be a hopeless task. It may be several lifetimes before we locate systems similar to ours. I had hoped to find some in my day."

"Don't be discouraged, professor. I'll start hitting Cygnus for you, and maybe we can find something there. Yesterday, for a minute, I thought I had something in Lyra. In the middle of the random noise, suddenly I came across what looked like 'dot dot, dot dot, dot-dot-dot-dot, dot dot dot, dot dot dot—' It was clear as crystal, and much louder than the noise, but then it lapsed into the usual nonsense. Pure chance at work, of course, but for a minute there, my hair stood on end. Well, it's all in the day's work. I'll just keep sending our stuff—"

"Of course," said the anthropologist, "there might be intelligent people in the sector you're working right now, but they might be like the Zuni."

"The Zuni?" asked Weber.

"The Zuni are a large village of American Indians who live in New Mexico. Right next to what used to be a big Indian reservation for the Navahos. A typical inward-looking culture. Now the Navahos, although they still keep their own language and religion, are an outward-looking culture, interested in the rest of the world. The Zuni are not. For them the boundaries of the world are the walls of Zuni village. They wouldn't bother to listen to a message from outer space, much less reply."

"Can't be very bright," said Joe.

"On the contrary. Some of them have made the highest scores on the Tromovich intelligence test that have ever been recorded. It's not a matter of intelligence, but the attitude of the culture."

"Well," said Joe, "let's hope most of the worlds of space have outward-looking cultures." He turned back to his transmitter.

Pehn found himself lying on a couch. He tried to get up, and felt his father's hands at his shoulders restraining him.

"Don't move, Pehn," said his mother. "You fainted, and they brought you home." She lifted his head, and let him sip a cup of the hot *akhlai*.

After a time his father addressed him with unusual kindness.

"We cannot delay any longer, Pehn. Your mother and sister and I are all agreed. You must undergo Ceremony. What is proper thing to do for you, I don't know, but Bidagha believes he can help you. This crisis in our world is making us all ill, and it is no wonder that you, being young, should suffer more acutely than rest of us."

Pehn tried to laugh. "Would you trust me to Bidagha?"

"Yes, I would. Although his views on world affairs are perverse and dangerous he is good Healer, and he has your best interests at heart."

"Your son is very ill, Lord Karn. If I am able to help him, would you be willing to consider possibility, at least, that wisdom of Healer is not confined to human body alone?"

Premier Karn brushed his hand across his eyes. In the last few days he had suddenly become an old man, and his mouth was drawn and tense. "I cannot tell, Bidagha. I am tired, and confused. I no longer seem to be sure what is true, and what is right."

Pehn opened his eyes to speak to Bidagha. "Do you think you can cure me, by Ceremony alone?"

"I have cured people who were much more ill than you are, but your case is serious because you have delayed so long. It may be that we should not rely

on ritual alone, and that it would be wiser to use knife."

Lord Karn gasped with horror. "Never! Have you lost faith in your own art?"

"Of course none of us like to use knife, since very few minds are skilled enough to control infection likely to follow. No, first we shall hold Ceremony just as our traditions counsel us. Tonight."

The muted chime of the visiphone interrupted them. The Premier touched his wrist and the gaunt face of Nautunal appeared in the dial.

"Signals have entered new stage, sir. Message coming at present states that Topaz planet wishes to visit any planet inhabited by intelligent race. They say they have phase space drive for spaceships, but in all their searching have not found even one inhabited world. They say they want to know that they are not alone in universe."

The Premier's face worked. "I cannot say what is right. Later. I will decide after Ceremony. Are you ready, Pehn?"

Pehn covered his face with his folded arms.

"All right," he whispered. "Sooner, better."

Deep in the valley north of the city, Cave yawned. For thousands of years its narrow mouth had been open to the Healers and the participants in the ceremonial ritual. The age of Cave was unknown, some said as old as the planet itself. Great rocks formed the inner walls, which ascended to a low domed ceiling, and occasionally a handful of gravel trickled down the walls to the bottom where a small stream still worked at hollowing out the stone.

At the back of Cave was the hearth, and across the floor were ancient stone benches waiting for the friends and family of the patient, for by tradition, only patient and Healer approached the hearth itself. The others, whose wills and hearts were to unite, for one brief night, to heal the sickness, sat apart in a broad circle, where they could see the ceremony, and the chant of their voices could float back to the ears of the sufferer.

As the sun set, Pehn was carried into Cave on a litter.

His father and mother, his sister, his father's collaborators in the government, and representatives of the whole community filed down the valley to the entrance where Bidagha stood, and each person, clad for this occasion only in a robe of animal fur, as he approached the opening extended his hands to show that he had removed his wrist band, and lifted his arms to show that no material product of modern technology was being taken inside to profane Cave. They all respected the ancient proverb, "What Immortals want new, they make new." Each one lit his candle at Bidagha's flame, and silently took his place in the circle.

Pehn had not been inside Cave since his early childhood, but it seemed a familiar place, since its description formed a part of many of Zenob's myths, and was part of all her history.

The age-old figures scratched on the wall and filled in with colored earths, had been made by his remote ancestors at a time when their only weapons were the bows and arrows pictured there, and the stone-tipped spears with which they hunted their game. In the flickering light of the fire he could recognize the lithe *toda*, and the great-tusked *khalmat*, animals which had been extinct for many ages. How vividly Old Ones had portrayed these animals, and the ritual of their hunts! The wood fire, which Bidagha had kindled with a primitive wooden drill, burned on the hearth, and above his head through a rift in the ceiling, Pehn could see a narrow band of sky and a sprinkling of stars.

"Keep your head pointed towards fire," said Bidagha, "and lie quiet. Ritual has no value unless we observe it strictly." He gave Pehn a warm potion from an earthenware cup, which made him feel sleepy.

Bidagha began to chant, his bass voice reverberating from wall to wall, each syllable a sonorous musical note which was answered at intervals by the watching group of well-wishers.

A wooden bowl filled with coarsely ground grain was passed from one person to another, and each one placed a few grains on his tongue, some on his forehead, and threw a token pinch of the flour over his left shoulder.

An hour passed, two; the stars above shifted their position, and still Bidagha chanted, never hesitating, never stumbling over the archaic words. Midnight

passed and the stars grew pale.

Through the roaring in his ears, Pehn heard the Healer kneel on the rock floor beside him; then he felt Bidagha's strong fingers on his shoulder.

"How is it with you, my son?"

Pehn groaned, unable to speak. The pain was not alleviated—it was greater then ever.

The soles of Bidagha's sandals scraped as he stood up again.

"Bring knife!" he called.

In his roaring darkness, Pehn stirred. Vaguely he sensed the murmuring of the watchers. Then someone else came near, and Bidagha's voice rose again. "Immortals, bless knife!"

Fingers pried open his jaw, probed at the misshapen gum, sending fiery flashes of agony into his brain. Then a hard edge of pain struck, cutting, releasing a flood of warm wetness in his mouth. Yet it all seemed to be happening far away.

He sensed Bidagha bending near once more. "Boy is going fast. Infection is deep."

Another voice: "Move him to experimental hospital?"

"He would not live to get there." A pause. "Go, bring forceps and bone knives. Hurry."

A long roaring darkness. Then new movement around where he lay: and a sudden voice that he dimly recognized as his father's.

"Stop! What is that tool in your hand?"

"A new device for extracting teeth," came Bidagha's calm, resonant voice: "with which we may save your son's life."

Shocked murmurs all over the hall, topped by his father's shout of outrage, "In Cave—in hands of Healer?"

Bidagha relied, "What Immortals want new, they make new! Here and now, in my hands, they end our years of darkness! Let Immortals confound me if I lie!"

The multitude in Cave roared their approval, and Premier Karn hesitated. He appeared to be struggling within himself. As the echoes died away, a pebble rolled from a ledge, dislodged by the sound, and fell at Bidagha's feet. A second pebble fell, and a boulder which had rested above the hearth for untold centuries shifted its position.

With a shout, Bidagha flung himself over Pehn's body as the boulder trembled and fell, crushing the life from the bodies of both men.

Dust rose, and a rumble began near the ceiling.

"Run!" cried Premier Karn. "Run for your lives!"

As the others ran from Cave, Lord Karn rushed to the huge rock lying upon his son, but he had no hope. Neither Pehn nor Bidagha would ever move again.

A trickle of sand pattered to the floor, and with a last backward glance Lord Karn ran from Cave. Boulders rained from the ceiling. The Premier had just reached the outside when a huge slab of rock crashed to the floor against the entrance. On the slope nearby, Pehn's mother and sister wept silently.

Lord Karn stood motionless a long while. At last he spoke.

"Cave is sealed," he said. "Let it never be opened again. Immortals have willed that my son should rest here forever, with impious Bidagha." Turning his face to the sky, he shook his fist at the bright spark of Topaz in the paling north. "So much for new things and foreign stars!" he said between his teeth. "This day's evil is enough."

They extinguished their candles and went slowly up the valley path towards the city.

Twenty-eight years later, on Earth, an astronomer comparing recent plates taken of the constellation Lyra noticed that Vega, its brightest star, had increased in brightness by a slight amount. The event was not especially remarkable—there are on the average, twenty-five novas reported every

year in our galaxy—but Vega was one of the stars to be visited during the next decade by one of the Survey ships now in mid-voyage.

"There's one place they won't have to stop, now," he said to a colleague, showing him the plates.

"I don't suppose it matters. What's one star, more or less, when they all turn out to be the same—no planets, or barren ones—no stopping place for man."

"I suppose you're right," said the astronomer, staring glumly at the waste immensity of the photograph in front of him.

*** END OF THE PROJECT GUTENBERG EBOOK A TOOTHACHE
ON ZENOB ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.